

PROJECT SIMULATION

AUDIO AMPLIFIER

TEAM

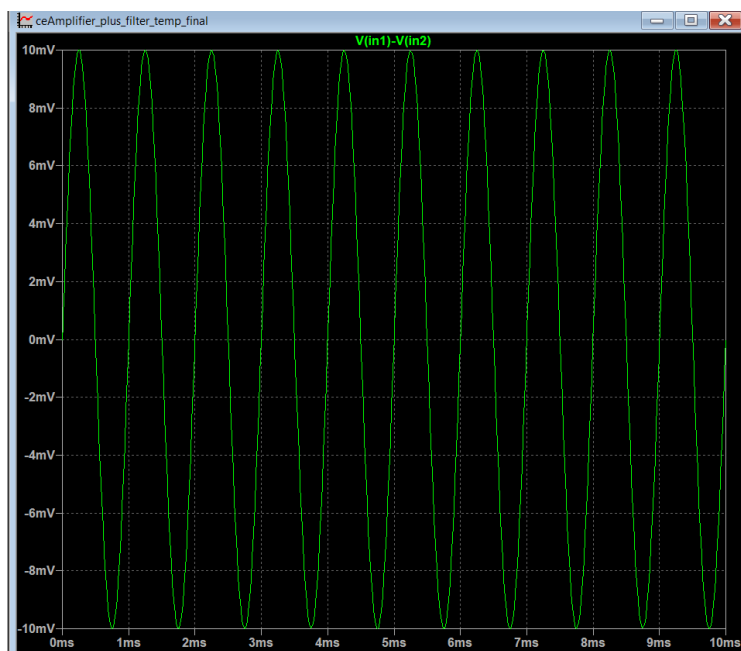
YOGESH KRISHNA – 2024102035

ANUMAY RAI – 2025122013

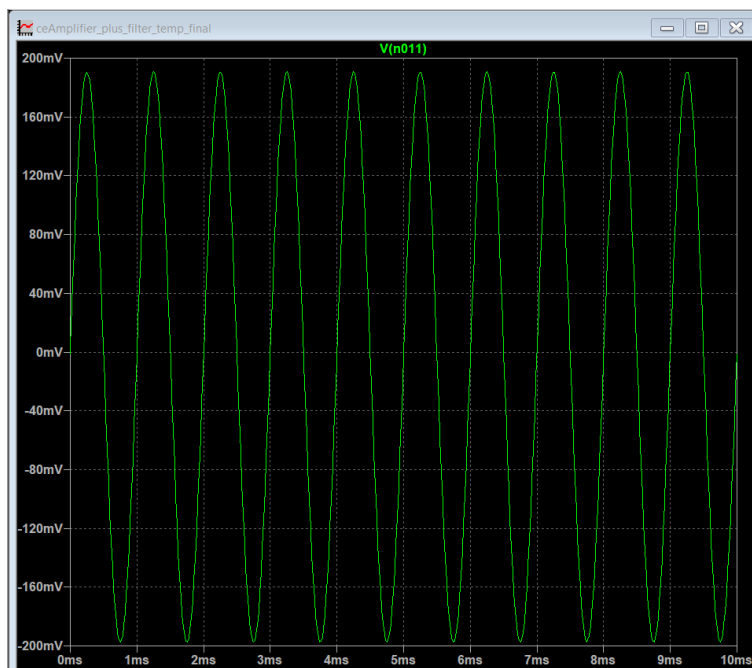
TABLE NUMBER – 10

ROOM – N117

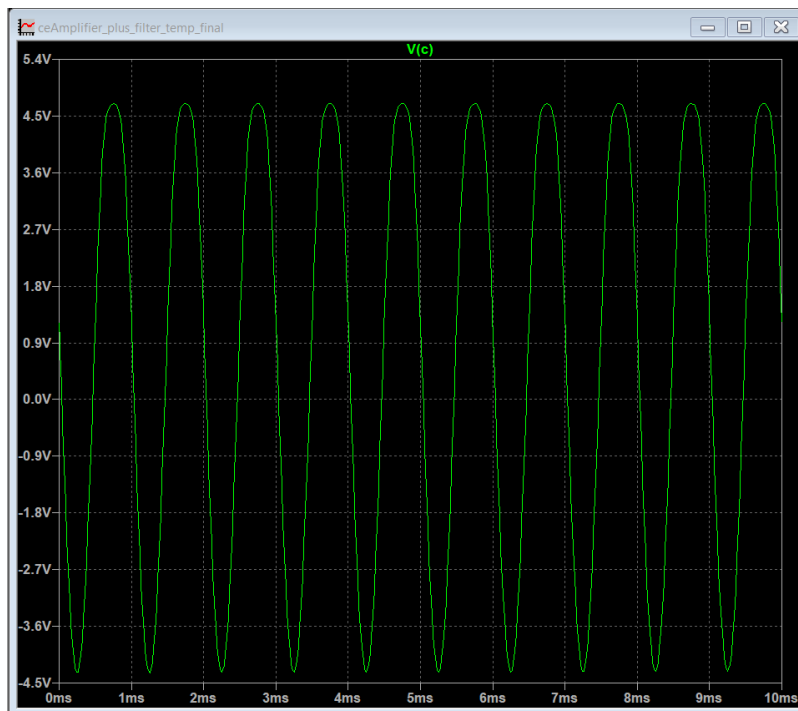
Input Voltage



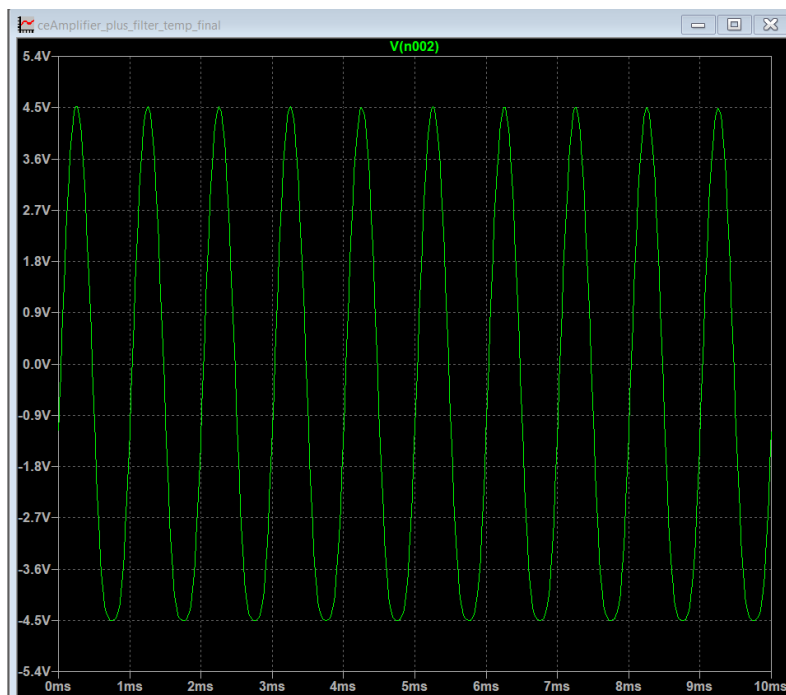
Output of Differential Circuit



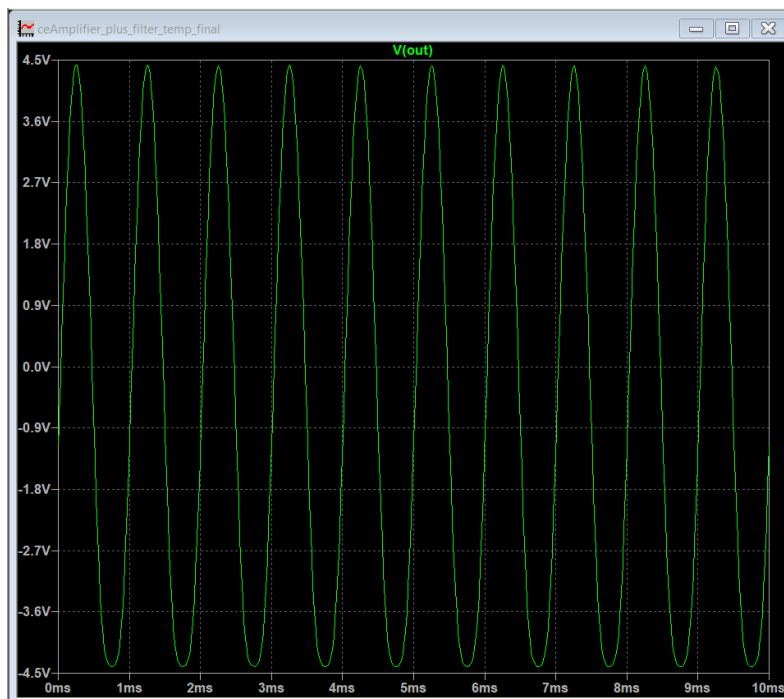
Output of CE Amplifier



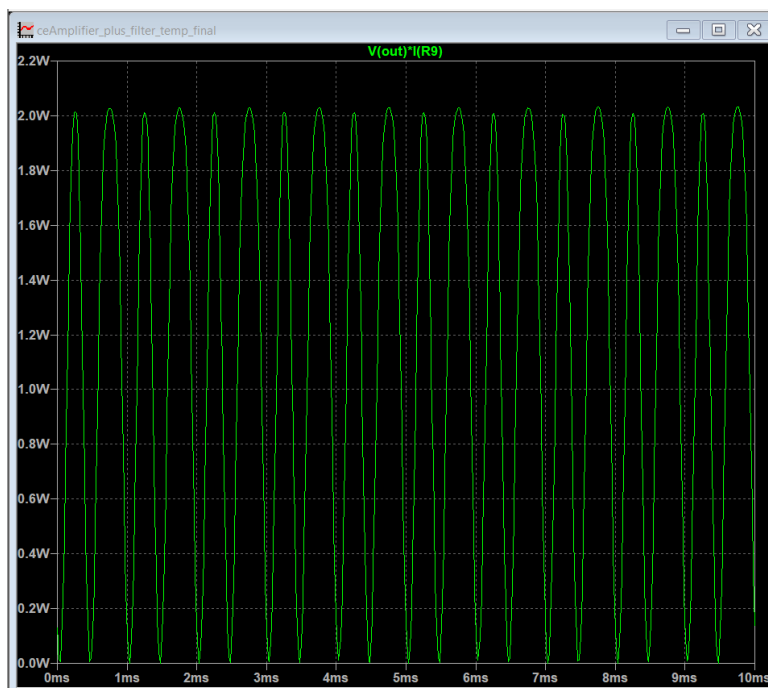
Output of Bandpass Filter



Output of Power Amplifier

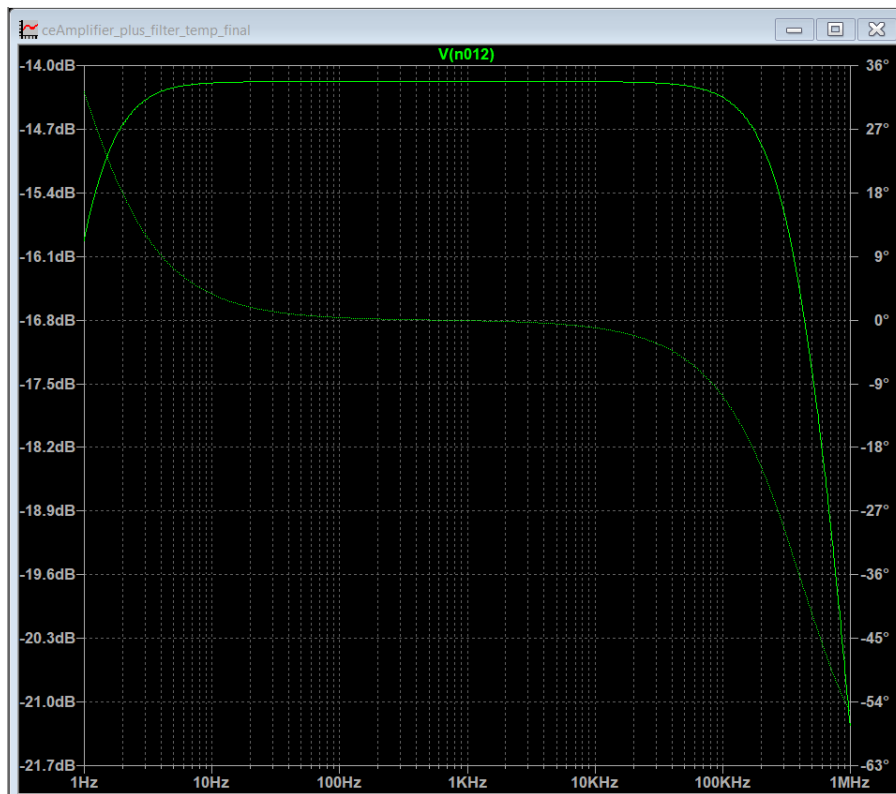


Final Power

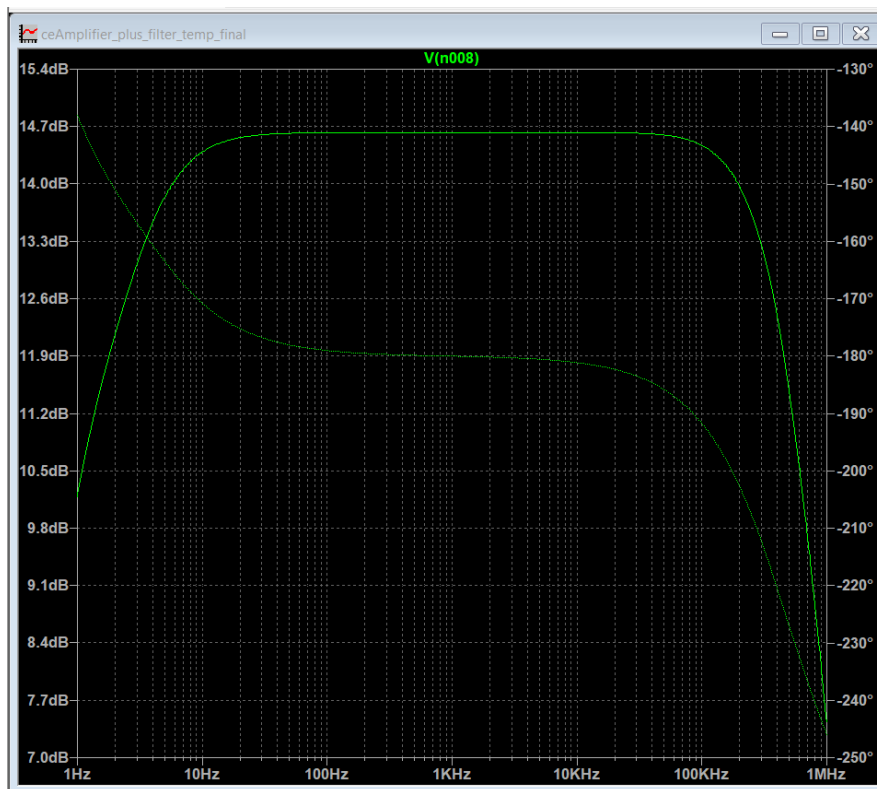


Bode Plots

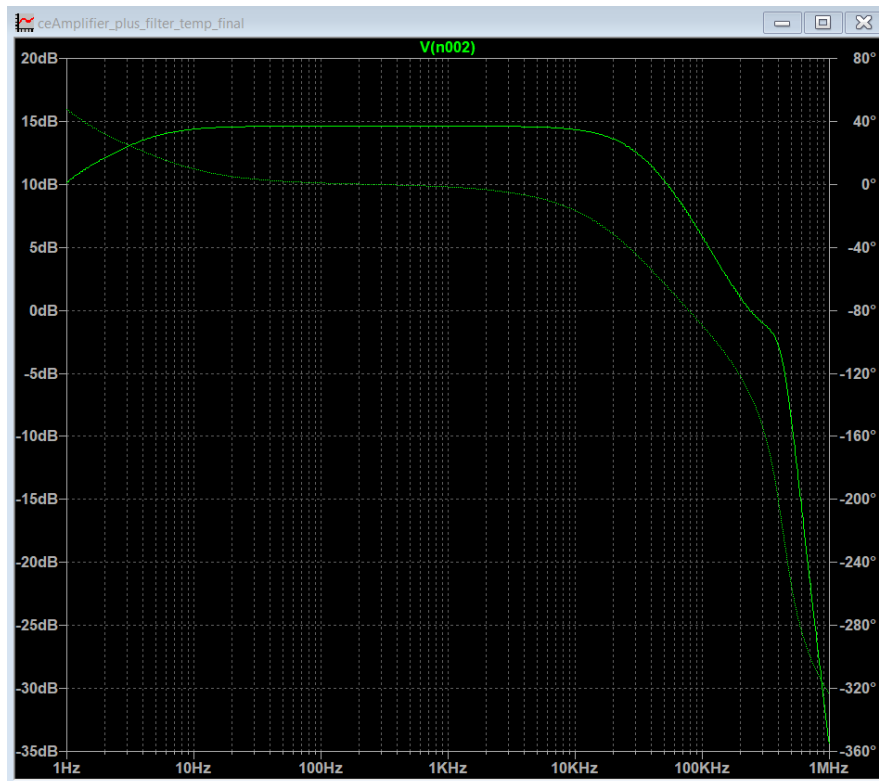
1. Differential Circuit



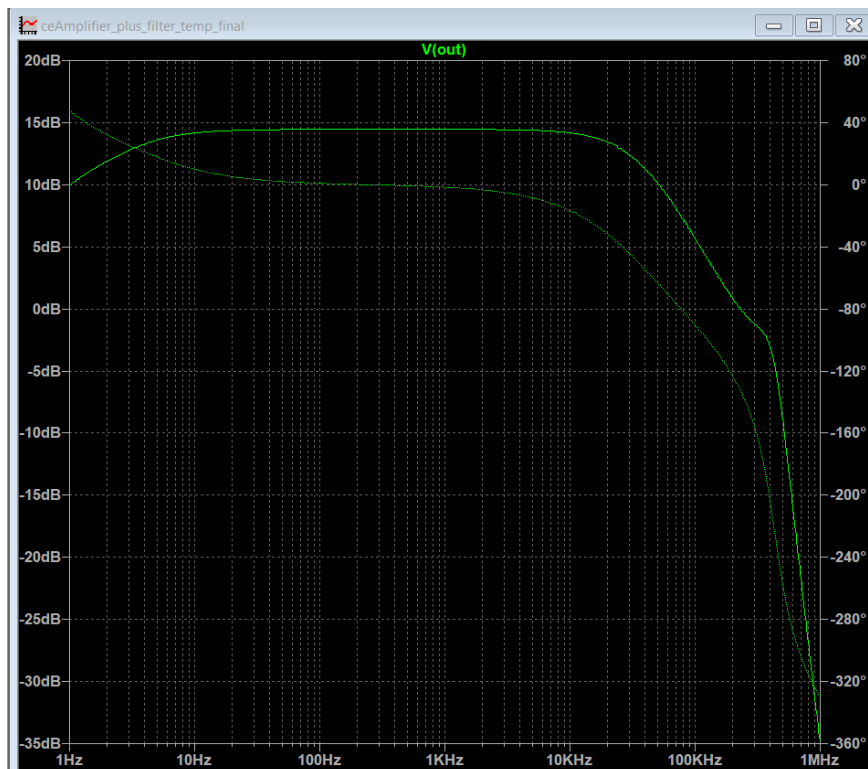
2. CE Amplifier



3. Bandstop Filter



4. Power Amplifier



Gain Stages

1. Pre- Amplifier Gain –
 - a) 20mV pk-pk at 1kHz = 19.2
 - b) 10mV pk-pk at 1KHz = 19.4
2. Amplifier Gain
 - a) 20mV pk-pk at 1KHz = 24.0
 - b) 10mV pk-pk at 1KHz = 26.8
3. Filter Gain
 - c) 20mV pk-pk at 1KHz = 1
 - d) 10mV pk-pk at 1KHz = 1
4. Amplifier Gain
 - e) 20mV pk-pk at 1KHz = 0.97
 - f) 10mV pk-pk at 1KHz = 0.98
5. Overall Gain
 - g) 20mV pk-pk at 1KHz = 449.5
 - h) 10mV pk-pk at 1KHz = 508.9
6. Overall Gain(Without Power Amplifier)
 - i) 20mV pk-pk at 1KHz = 459.5
 - j) 10mV pk-pk at 1KHz = 519.3

Parameters

1. Common Mode Rejection Ratio(CMRR) = 30.4dB
2. Slew Rate = 0.057 V/ μ s
3. Total Harmonic Distortion (THD)
 - a) 20mV pk-pk at 1kHz = 11.862086%
 - b) 10mV pk-pk at 1kHz = 3.893069%
4. Stability Factor (S) = 11.1048